D07 · Warehouse KPIs, Metrics & Governance

05 · Warehouse safety, compliance & ESG in logistics operations

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1 WHAT IT IS

Warehouse safety and compliance protect people and the operation through risk controls, training and mandatory standards — in Mexico, STPS norms such as NOM-006-STPS (materials handling and storage) and NOM-009-STPS (work at heights) — while ESG adds energy, emissions, waste and working conditions.

2 WHY IT MATTERS

Warehouses combine moving equipment, people on foot, heights and heavy loads. **An accident has a human cost first, and then legal, financial and operational costs that no productivity gain compensates.** Safety improves most when it is managed proactively — observing and correcting unsafe conditions — and made visible to leadership with the same weight as cost and service.

3 HOW IT IS DONE

1. **Assess risks by zone.** Forklift traffic, racking, picking ergonomics, docks and hazardous materials.
2. **Separate people and equipment.** Marked walkways, barriers, speed limits and certified operators.
3. **Run a proactive observation program.** Regular observations, corrected conditions and leading indicators.
4. **Report safety and ESG at leadership level.** TRIR and environmental metrics on the same dashboard as cost and service.

IN PRACTICE

Safety culture is built with proactive observations — not incident investigations

A safety observation program that observes and corrects unsafe behaviors before they cause an accident is more effective than root cause analysis of accidents that have already occurred.

Compliance with industrial safety NOM standards in the warehouse is not optional

NOM-006-STPS (material handling and storage) and NOM-009-STPS (work at heights) are mandatory compliance in Mexico. The STPS conducts inspections that can result in fines and operation suspension.

Migration from combustion to electric forklifts has financial ROI beyond ESG

Electric forklifts have significantly lower operating cost than gas or diesel — in addition to eliminating indoor emissions. Typical ROI: 3–5 years.

Include warehouse safety in the S&OP dashboard as an organizational culture KPI

Warehouse TRIR in the S&OP dashboard converts safety into a management-visible indicator — with the same priority as fill rate and cost per order.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Warehouse safety program — DC with a TRIR of 3.2 — Program based on NOM-006-STPS-2014 and NOM-009-STPS (work at heights).

Program element	Implementation	Result
Physical separation of pedestrians and forklifts (zebra crossings + barriers)	Month 1	0 run-over incidents in 12 months post-implementation vs. 2 in prior 12 months

Program element	Implementation	Result
Certification of all forklift operators (NOM-006)	Month 2–3	100% of operators certified vs. 62% before the program
Safety observation program (STOP)	Month 3	42 observations/month average · 12 unsafe conditions corrected proactively

Result: TRIR: 3.2 -> 0.8 recordable incidents per 200,000 hours worked (-75%). 0 fatal or disabling incidents in the 12-month program. The observation program was the highest-impact initiative — converting safety from reactive to proactive.

HOW IT IS MEASURED

TRIR — Total Recordable Incident Rate

TRIR — Total Recordable Incident Rate(Number of recordable incidents × 200,000) / Total hours worked

Benchmark: <1.0 TRIR in mature warehouse operations · compare with official statistics for warehousing (BLS in the US, IMSS in Mexico) · 0 is the ideal target A TRIR in an upward trend is the most urgent possible indicator in warehouse operations. Incidents have human, legal, financial, and reputational consequences that no productivity dashboard can compensate.

Warehouse Carbon Intensity (tCO per unit of warehouse throughput)

Warehouse Carbon IntensityTotal tCO of warehouse operation / Warehouse throughput (orders or units)

Benchmark: 5–10% year-on-year reduction in operations with active decarbonization programs Migration from combustion to electric forklifts and LED lighting are the highest-impact initiatives for warehouse carbon footprint.

WHAT YOU WOULD USE

Safety & ESG

STPS / IMSS — Module: Warehouse Safety Compliance + NOM StandardsThe reference regulatory bodies for industrial safety in Mexico.

Intelex / Cority — Module: EHS Management PlatformHealth, safety, and environment (EHS) management platforms for incident tracking, safety observations, and carbon footprint reporting.

4 SO WHAT

Make safety a leadership metric. Comply with STPS norms, separate people from equipment, observe proactively and put TRIR and ESG on the same dashboard as cost and service.

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